

Day 28 Short Written Final

Geometry Summer Credit | Course Synthesis

Name:	Date:	Score:
-------	-------	--------

Directions

Show work for written items. This final emphasizes reasoning, formula choice, and readiness for Algebra 2/Trigonometry.

1. The distance between (0,0) and (8,6) is:
A 7 B 10 C 14 D 48

2. A line perpendicular to slope $\frac{2}{3}$ has slope:
A $\frac{2}{3}$ B $-\frac{2}{3}$ C $\frac{3}{2}$ D $-\frac{3}{2}$

3. If two similar figures have scale factor 5, their area scale factor is:
A 5 B 10 C 25 D 125

4. The circle $(x + 2)^2 + (y - 4)^2 = 49$ has center:
A (2,-4) B (-2,4) C (-2,-4) D (2,4)

5. Tangent in a right triangle is:
A opposite/hypotenuse B adjacent/hypotenuse C opposite/adjacent D hypotenuse/adjacent

6. Write the equation of a circle with center (5, -3) and radius 4.

7. A circle has radius 12 and central angle 90 degrees. Find the sector area.

Day 28 Short Written Final

Continued

8.	A right triangle has angle 37 degrees and adjacent side 20. Find the hypotenuse to the nearest tenth.
----	---

9.	Find the volume of a cone with radius 6 and height 9 in terms of pi.
----	--

10.	A package has volume 280 cm^3 and density 7 g/cm^3 . It must weigh less than 2000 g. Does it meet the constraint? Explain.
-----	--

11.	Describe one connection between Geometry and Algebra 2 or Trigonometry.
-----	---

12.	Error analysis: A student finds arc length when the problem asks for sector area. Explain the error and give the correct formula.
-----	---